

Project title-1:

Real-Time Vehicle Fuel Distribution and Black-Market Theft Prevention Platform

Introduction

Fuel is a critical resource for Ethiopia's transportation sector, government institutions, public transport services, and fleet-based organizations. Efficient fuel management is essential to ensure operational continuity, economic stability, and public service delivery. However, the current fuel allocation and distribution processes largely rely on manual record-keeping systems and weak control mechanisms.

Statement of the Problem

Fuel mismanagement and theft are major challenges in Ethiopia. Manual records and weak control systems create opportunities for theft, over-dispensing, and illegal diversion of fuel. Fuel is frequently redirected and resold on the black market at inflated prices, increasing transportation costs and causing significant financial losses for institutions.

In some cases, fuel stations, corrupt officials, and black-market actors cooperate, worsening the situation and making fuel less accessible to legitimate drivers. Therefore, a critical need for a secure, automated, and transparent digital system that ensures fuel is dispensed strictly to authorized drivers and that every transaction is recorded and monitored in real time.

Objectives

The project aims to develop a real-time digital platform that:

- Implements driver-based fuel allocation with daily limits
- Uses QR code authentication for secure verification
- Monitors fuel station stock levels in real time
- Records all transactions in a centralized database
- Provides an administrative dashboard for oversight
- Reduces fuel diversion through automation and transparency

Significance of the project

The platform will improve transparency, accountability, and efficiency in Ethiopia's fuel distribution system. It will reduce corruption, protect drivers from black-market exploitation, support better governance, and provide reliable data for decision-making. If successful, it can serve as a scalable national solution for modernizing fuel management and reducing systemic fuel theft.

Project title-2 : **Integrated Digital Payment and Fare Management System for Shared Urban Transport**

Introduction

Shared transport services such as minibuses and shared taxis play a vital role in daily mobility and economic activity. However, fare collection in these services remains largely cash-based, creating operational inefficiencies, limiting transparency, and making regulatory enforcement difficult.

Beyond digital payments, the system also provides a centralized monitoring platform that enables government authorities to regulate taxi tariffs and ensure that vehicles operate only on their officially assigned routes and stations. This integration enhances transparency, regulatory compliance, and public trust in the shared transport sector.

Statement of the problem

Despite the widespread use of minibuses and shared taxis, fare collection remains dependent on physical cash transactions. This cash-based system leads to frequent money loss, lack of exact change, disputes between drivers and passengers, and limited financial transparency.

In addition, there is limited regulatory control over fare pricing and route compliance. Taxi drivers may arbitrarily increase tariffs, overcharge passengers, or deviate from their authorized stations and designated routes. The absence of a centralized monitoring and enforcement system makes it difficult for authorities to control pricing, enforce transport regulations, and ensure fair service delivery.

Objectives

The project aims to:

- Replace cash-based fare collection with a QR-based digital payment system
- Include both smartphone and non-smartphone passengers
- Enable secure, real-time fare deduction and transaction recording
- Reduce disputes between drivers and passengers
- Provide transparent digital records for accountability

Significance of the project

The system improves transparency, operational efficiency, and financial accountability in shared transport. It ensures secure and convenient payments, reduces cash handling, minimizes disputes, and provides a reliable digital record for all fare transactions. Importantly, the platform empowers government authorities to regulate transport tariffs effectively and ensure compliance with approved routes and station assignments.

Project title-3:

Digital Promotion and Channel Aggregation Platform for Ethiopia

Introduction

The rapid growth of Telegram and other social media platforms has significantly transformed digital marketing practices in Ethiopia. Many businesses including hotels, restaurants, e-commerce sellers, and content creators now rely heavily on Telegram channels and social networks to promote their products and services.

As digital engagement continues to expand, the need for structured and organized promotional systems becomes increasingly important. A well-designed platform that centralizes and categorizes promotional channels can enhance visibility, improve customer access, and strengthen digital marketing efficiency.

Statement of the Problem

Despite the increasing reliance on Telegram and social media for business promotion, there is currently no centralized, structured, or location-based system in Ethiopia that aggregates and organizes these promotional channels.

As a result, users struggle to find verified businesses by category or geographic location. Promotions are scattered, unstructured, and difficult to navigate. Businesses also lack effective tools to manage visibility, target appropriate audiences, and enhance customer engagement.

Objectives

The project aims to design and develop a centralized digital promotion platform that:

- Allows users to browse and filter channels by category and location
- Provides a secure web-based admin portal for content and category management
- Integrates Telegram channel linking for direct access
- Supports time-based promotion management (e.g., monthly plans)

Significance of the project

The platform will improve digital service discovery and structured promotion in Ethiopia. It will help users easily access verified businesses, enable companies to reach targeted audiences more effectively, and support content creators in increasing visibility. Additionally, it will strengthen Ethiopia's digital marketing ecosystem and provide a scalable foundation for future expansion, including analytics and automated payment systems.